ANAND NAGARAJ

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Research & Technical Profile: Master's student in Design and Test of Integrated Circuit at Technische Universät Chemnitz, Germany with 3+ years of professional industry experience as an RTL & VLSI Engineer. Specialized in **RISC-V processor architecture**, **memory subsystems**, **Network-on-Chip (NoC)** interconnects, and **high-speed serial protocols (PCIe Gen3/4)**. Experienced in digital design, SystemVerilog/UVM verification, FPGA bring-up, and open-source EDA tapeouts.

EDUCATION

M.Sc. in Design and Test of Integrated Circuits (DTIC) — Apr 2026 – Present
Technische Universität Chemnitz, Germany — Chemnitz, Germany
Focus: Advanced Digital IC Design, Design & Verifiucation of digital systems, VHDL-AMS Modeling, Heterogeneous Systems, Elements of IC

B.E. in Electronics & Telecommunication Engineering — Aug 2018 – Apr 2022
Visvesvaraya Technological University (BMSIT), Bengaluru, India — Bengaluru, India
Focus: Hardware description language, Digital System Design, Microprocessors, Computer architecture, CMOS

PROFESSIONAL EXPERIENCE

VLSI Design Engineer (Founding Team Member) — Nov 2024 – Feb 2026
IndusIP Labs Pvt Ltd — Bengaluru, India

- **Network-on-Chip (NoC) IP:** Designed, verified, and integrated synthesizable RTL modules for high-throughput NoC routers, including round-robin arbiters, multi-flit input/output buffers, crossbar matrix switches, and routing logic.
- **RISC-V Microarchitecture & Extensions:** Analyzed open-source Ibex and SweRV RISC-V cores; implemented custom instruction extensions in SystemVerilog and verified functional correctness against the Spike Golden ISS model.
- **EDA Automation & Datasets:** Curated and verified extensive RTL datasets (FSMs, AMBA AHB/AXI protocol bridges, memory controllers) to train AI/LLM models for automated RTL generation, linting, and simulation flows.
- **Technical Mentorship:** Guided junior engineers and interns through the full digital implementation pipeline towards GDS.

RTL Design Engineer — Sep 2022 – Oct 2024
Sinergia Media Labs — Bengaluru, India

- **PCIe Transaction Layer Design:** Implemented synthesizable RTL for the PCIe Transaction Layer, managing generation and processing of 32-bit/64-bit Request and Completion TLPs with integrated ECRC logic.
- **Flow Control & Buffering:** Architected credit-based flow control tracking credit consumption and replenishment, preventing buffer overruns between requester and completer nodes under burst traffic.
- **FPGA Prototyping & Bring-Up:** Executed on-chip validation and debug on Xilinx FPGAs using Vivado ILA/VIO and ModelSim, closing static timing (STA) constraints and validating PCIe protocol compliance.
- **ChipGPT Framework:** Co-developed an automated RTL generation and verification workflow integrating Python with EDA tools for linting, synthesis, simulation, and static timing analysis.

PUBLICATIONS

- "**Peripheral Shared Memory over PCI Express Link**," in *Proc. 7th International Conference on Innovative Data Communication Technologies and Application (ICIDCA)*, 2025.
  Focus: High-throughput low-latency peripheral memory sharing mechanisms and protocol-level transactions over PCIe Gen3/4 physical and transaction layers. [ResearchGate Publication]

TECHNICAL PROJECTS & OPEN-SOURCE SILICON

Moola-V (MR5): Synthesizable 5-Stage Pipelined RV32I Core [GitHub: moola_riscv_core]
SystemVerilog, RISCOF, Spike ISS, Cocotb, Icarus Verilog, GTKWave

- Architected a 5-stage in-order pipelined 32-bit RISC-V (RV32I) core from scratch (Fetch, Decode, Execute, Memory, Write-Back) supporting all 40 base integer instructions across R, I, S, B, U, and J formats.
- Built full RAW operand forwarding (EX-to-EX, MEM-to-EX) with link address handling, load-use interlocking with cycle-accurate bubble injection, and two-instruction branch/jump pipeline flush logic.
- Achieved **100% compliance (38/38 tests passing)** in the official RISC-V Architectural Test Framework (**RISCOF**) via Spike co-simulation; validated corner cases with Cocotb lock-step Python ISS co-simulation.

TT Moola Dual-Port Synchronous SRAM Macro [GitHub: tt_moola_dp_sram]
Verilog, SkyWater 130nm, OpenLane 2, Tiny Tapeout, Cocotb

- Designed and synthesized a 16×8 parameterized synchronous dual-port SRAM macro module for integration on the SkyWater 130nm process node targeting the Tiny Tapeout ASIC framework.
- Implemented byte-enable write strobes (`wstrb[3:0]`), synchronous registered read with 1-cycle latency, and address collision arbitration for concurrent read/write ports.

- Verified memory array integrity, byte masking, and timing constraints using Cocotb testbenches and automated OpenROAD/OpenLane physical verification flows.

AMBA AHB-to-APB Bridge IP Design & UVM Verification *Verilog, SystemVerilog, UVM*

- Designed an AMBA-compliant bridge converting pipelined high-bandwidth AHB transfers to non-pipelined single-cycle APB peripheral handshakes with wait-state and error response logic.
- Built a modular UVM environment featuring AHB master and APB slave UVCs; deployed constrained-random sequences and SVA assertions, achieving 100% functional and code coverage closure.

CERTIFICATIONS & SPECIALIZED TRAINING

Advanced VLSI Design and Verification (6-Month Technical Vocational Program) Apr 2022 – Sep 2022
Maven Silicon Pvt Ltd, Bengaluru, India

- In-depth industrial curriculum covering SystemVerilog, UVM testbench architecture, SystemVerilog Assertions (SVA), AMBA bus protocols (AXI/AHB/APB), coverage closure, and ASIC synthesis.

TECHNICAL SKILLS

Hardware Description: SystemVerilog, Verilog, VHDL, VHDL-AMS
Verification & Methodologies: UVM, SystemVerilog Assertions (SVA), Constrained Random Verification, Cocotb, RISCOF
Architecture & Protocols: RISC-V (RV32I, Ibex, SweRV), Memory Subsystems (SRAM, DRAM, Caches), PCIe Gen3/4, AMBA (AXI4, AHB, APB)
EDA & Tools: Xilinx Vivado, ModelSim, QuestaSim, Cadence Genus/Innovus, Synopsys Design Compiler, OpenLane, OpenROAD, GTKWave
Languages & Scripting: Python, C/C++, Perl, Bash, Linux Environment

SCHOLARSHIPS & HONORS

- **Academic Merit Recognition & Tuition Support:** Awarded institutional scholarship recognition during undergraduate studies at VTU/BMSIT for top-percentile academic standing.
- **State Engineering Entrance Distinction:** Top-tier ranking in competitive pre-university engineering examinations.

SOCIAL RESPONSIBILITY & CIVIC ENGAGEMENT

- **General Elections 2018 Civic Volunteer:** Appointed civil volunteer by local authorities and police administration to coordinate logistics, direct voters, and facilitate accessible voting for elderly citizens.
- **AICTE Rural Digitalization Outreach:** Led demographic village surveys and organized digital literacy drives and STEM higher-education guidance workshops for rural students.

EXTRACURRICULARS, LEADERSHIP & SPORTS

- **Collegiate Sports Leadership:** Organized and managed collegiate-level cricket tournaments; master of ceremonies for university tech fests.
- **Athletics:** District-level runner-up in U-19 Kabaddi championship; competitive cricketer; distance runner and alpine trail hiker.

LANGUAGES

English: Fluent (C1 level) | **German:** Elementary (A2 Level)
Kannada: Native Proficiency | **Hindi:** Professional Working Proficiency